Multimedia Signal Processing - 2nd Module

February, 12 2019

1. **Question (7 pts)**
   Consider two discrete memoryless sources X and Y that emit symbols from the alphabets $\{x_1, x_2, x_3, x_4\}$ and $\{y_1, y_2\}$, respectively. The joint p.m.f. is

	y_1	y_2
x_1	0.05	0.15
x_2	0.35	0.05
x_3	0.2	0.1
x_4	0.05	0.05

   (a) Compute the probability mass functions of symbols of X and Y, separately.

   (b) Design the Huffman code to encode single symbols of X.

   (c) Considering point (b), compute: (i) the average code length of the obtained Huffman code; (ii) the average code length of the Shannon code, and; (iii) the theoretical bound.

   (d) Design the Huffman code that jointly encodes one symbol of X and one symbol of Y.

   (e) Considering point (d), compute again: (i) the average code length of the obtained Huffman code; (ii) the average code length of the Shannon code, and; (iii) the theoretical bound.

   (f) Compute the mutual information $I(X, Y)$.

2. **Question (6 pts)**
   Explain how to encode and decode a video using a hybrid DPCM-transform coding scheme. Clearly name each block and explain what it does. Clearly separate the encoding and decoding steps.

3. **Question (5 pts)**
   Explain what a Lloyd-Max quantizer is and how it works. Highlight the main difference with the optimum uniform scalar quantizer. Report an example in which Lloyd-Max quantization is not useful.

4. **Questions (5 pts - each answer can be either TRUE or FALSE - Incorrect answers give you -1 points, unanswered 0 points, correct answers 1 point - Provide a brief comment when in doubt)**

   - [T] [F] Given a fixed distortion, transform coding always provides better compression rate than DPCM.
   - [T] [F] Given a speech coding scheme based on DPCM, the long-term predictor helps particularly in coding pitched sounds with periodicities.
   - [T] [F] Given two sources X and Y characterized by finite element alphabets, mutual information $I(X, Y)$ is strictly positive.
   - [T] [F] Given a video coding scheme, the decoder does not need to know which motion estimation algorithm has been used by the encoder.
   - [T] [F] Considering perceptual audio coding, a tone can mask a noise, but a noise cannot mask a tone.

5. **Question (Matlab - 10pts)**

   Load the build-in 'pears.png' image file using the command `imread('pears.png')`.

   (a) Crop the top-left 256×256 pixels of the image and display it in full color (use the cropped version hereinafter).

   (b) Convert to grayscale and display the image again.

   (c) Quantize the grayscale image with $\{2, 3, 4, \ldots, 8\}$ bit/symbol and plot the SNR vs. rate. Compute the rate as the entropy of the quantized signal.

   (d) Apply a 2D-DCT block-wise transform coding scheme: split the image into 8×8 pixels blocks; for each block compute the 2D-DCT transform; quantize the block using 4 bit/symbol; reconstruct the image and display it.

   (e) Considering this last coding technique, compute the SNR and the rate. As rate use the entropy of the quantized DCT cofficients.

   (f) Is the DCT-based quantization better than the PCM quantization above? Why?